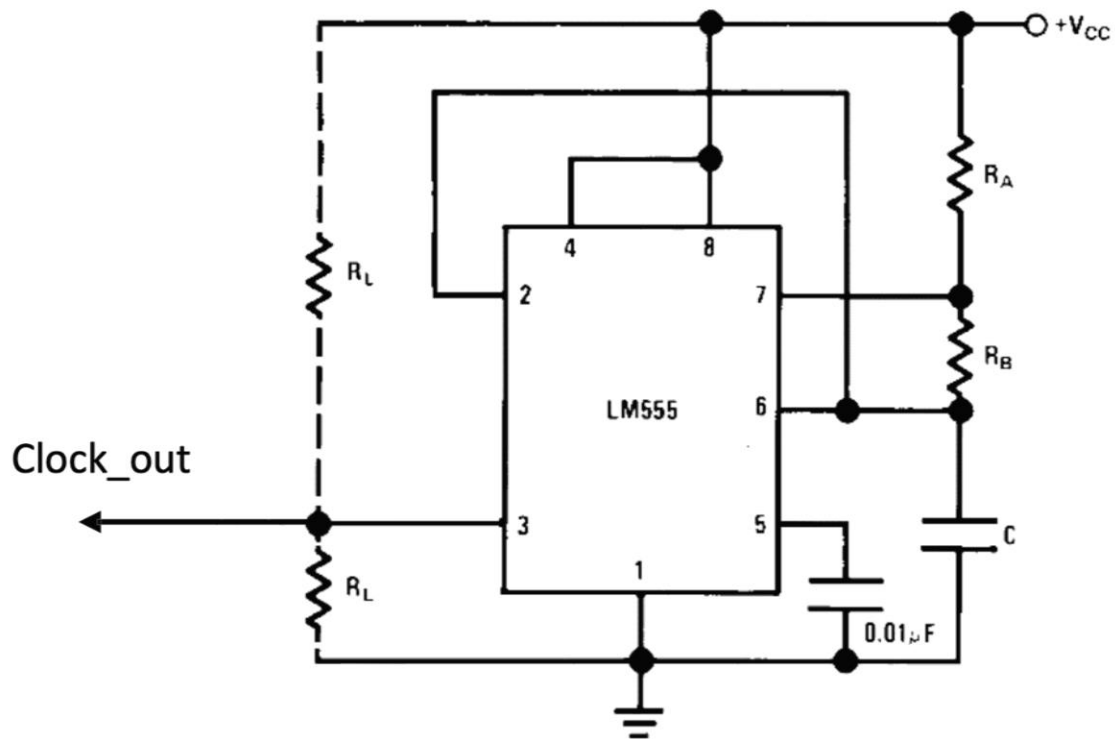


Clockgen Lab Report

ECE NTUA 2025-26

Digital Systems Laboratory



Εικόνα 1: Το ολοκληρωμένο κύκλωμα 555 σε ασταθή λειτουργία

Due to 16/03

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Theory Section

(Mathematical and Logical Operation of the Circuit)

Introduction

During the first laboratory exercise, an analog clock circuit was implemented using the LM555 integrated circuit in astable mode. The purpose of the circuit is to generate a periodic square-wave pulse, which is used as a timing signal for the broader digital circuit of the laboratory project. The circuit output reaches two distinct voltage levels, approximately 0 V and 5 V, and alternates periodically between them, thus producing a square waveform.

The periodic behavior of the signal is determined by the passive elements of the circuit, namely the resistors R_1 , R_2 , and the capacitor C , while the LM555 acts as the active element that controls the charging and discharging of the capacitor and ultimately produces the output pulse.

Internal Logical Structure of the LM555

Internally, the LM555 contains:

- a voltage divider consisting of three equal resistors-5 Ohms each,
- two (voltage) comparators,

- an SR latch,
- and a discharge transistor connected to pin 7.

The voltage divider divides the supply voltage $V_{cc} = 5\text{ V}$ into the following reference levels:

- 0 V
- $1/3 V_{cc} \approx 1.67\text{ V}$
- $2/3 V_{cc} \approx 3.33\text{ V}$
- $V_{cc} = 5\text{ V}$

The levels $1/3 V_{cc}$ and $2/3 V_{cc}$ are used as the operating thresholds of the two comparators.

In astable mode:(these are described as implemented later in this report)

- pins 2 and 6 are connected together,
- pin 4 (reset) is permanently connected to V_{cc} so that no external reset is triggered,
- pin 5 is connected to ground through a $0.01\text{ }\mu\text{F}$ capacitor in order to reduce noise,
- pin 7 is used for discharging the capacitor,
- and pin 3 is the clock output (Clock_out).

Timelapse of Circuit Operation

When the circuit is just being powered, the capacitor is initially considered discharged or only slightly charged. Therefore, the voltage at pins 2 and 6 is lower than the threshold $1/3 V_{cc} \approx 1.67\text{ V}$. At that moment, the lower comparator is activated and sends a Set signal to the SR latch.

The Set state of the latch leads to the following:

- the output at pin 3 switches to logical HIGH (approximately +5 V),
- the discharge transistor at pin 7 remains OFF.

At this stage, the capacitor charges through the resistors $R_1 + R_2$.

As the capacitor charges, its voltage increases gradually. When the capacitor voltage exceeds the level $2/3 V_{cc} \approx 3.33$ V, the upper comparator is activated. This comparator sends a Reset signal to the latch.

The Reset state of the latch leads to the following:

- the output at pin 3 switches to logical LOW (approximately 0 V),
- the discharge transistor at pin 7 turns ON.

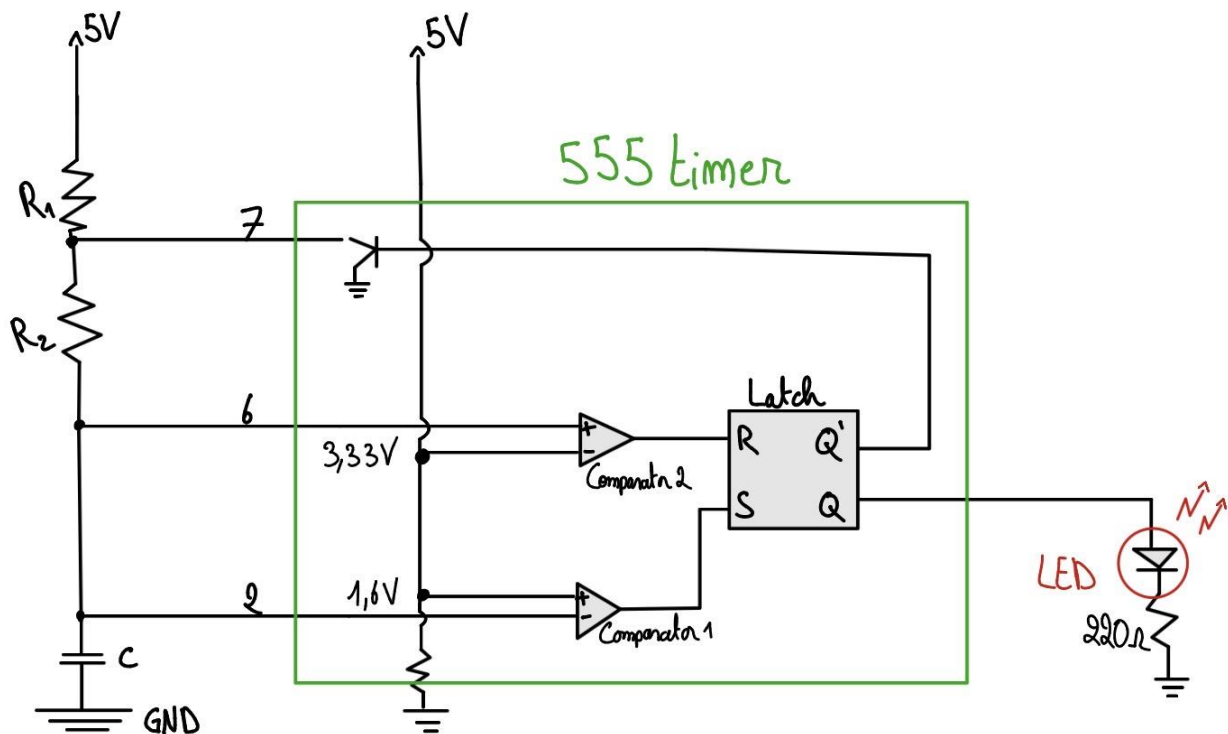
At this point, the capacitor begins to discharge, no longer through both resistors, but only through R_2 , since pin 7 is internally connected to ground through the discharge transistor. The discharge path is:

$C \rightarrow R_2 \rightarrow \text{pin } 7 \rightarrow \text{transistor} \rightarrow \text{GND}$ $\rightarrow R_2 \rightarrow \text{pin } 7 \rightarrow \text{transistor} \rightarrow \text{GND}$

During discharging, the capacitor voltage decreases. When it becomes lower than $1/3 V_{cc} \approx 1.67$ V, the lower comparator is activated again and sends a new Set signal to the latch. The discharge transistor is turned off, the output becomes HIGH again, and the charging cycle starts once more.

Thus, the circuit operates continuously and produces a periodic square waveform at its output.

The following image contains a handwritten version of the circuit that helped us visualize the process:



Voltage Waveforms

The output voltage V_{out} of the circuit is a square pulse that alternates between approximately **0 V** and **5 V**.

Obviously, the voltage across the capacitor V_C is not square. The capacitor:

- charges from approximately **1.67 V** to **3.33 V**,
- and then discharges from approximately **3.33 V** back down to **1.67 V**.

Therefore, the capacitor voltage follows the characteristic charging–discharging curve and oscillates between the two threshold levels of the integrated circuit.

The shape of the output waveform is determined by the charging and discharging durations of the capacitor, that is, by the values of **R_1** , **R_2** , and **C** .

Mathematical Relations

For the LM555 circuit operating in astable mode, the Texas Instr. manual's relationships are:

Time during which the output is HIGH

$$t_{HIGH} = 0.693 \cdot (R_1 + R_2) \cdot C$$

Time during which the output is LOW

$$t_{LOW} = 0.693 \cdot R_2 \cdot C$$

Oscillation period

$$T = t_{\text{HIGH}} + t_{\text{LOW}} \text{ therefore: } T = 0.693 \cdot (R_1 + 2R_2) \cdot C$$

Frequency

$$f = 1/T \text{ or equivalently: } f = 1.44 / (R_1 + 2R_2) \cdot C$$

Duty Cycle

The duty cycle is defined as the percentage of time during which the output remains at logical **HIGH** within one complete period: $D = (t_{\text{High}}/T)100\%$

Conclusion of the Theoretical Part

The analog clock circuit based on the **LM555** in astable mode is a simple yet highly effective periodic pulse generator. Its operation is based on the continuous charging and discharging of the capacitor between the voltage levels **1/3 Vcc** and **2/3 Vcc** through the resistors **R₁** and **R₂**, while the internal comparator and latch system of the LM555 converts this process into a clean square pulse at the output. By selecting the component values appropriately, it is possible to achieve the desired frequency and a nearly symmetrical duty cycle, allowing the circuit to be used reliably as an external

timing source for digital systems. Our element selection and circuit implementation is thoroughly explained in section 2: Laboratory Implementation, as follows:

Laboratory Section

(Implementation of the Circuit: two cases of the clockgen circuit)

1 – Component Selection and Circuit Assembly

This section covers the breadboard construction and the analysis of the signals captured via the oscilloscope.

To meet the lab requirements, as we explained on the 1st section, we configured the LM555 timer in astable mode. The circuit's operation is based on the continuous charge and discharge cycles of a capacitor through a resistor network. Before building the circuit, we calculated the values for R_A , R_B , and C to hit two specific targets: a visible period (roughly 1.5s) and a duty cycle as close to 50% as possible.

According to the datasheet, the period is governed by the following formula:

$$T=0.693 \cdot (R_A + 2R_B) \cdot C$$

- The Capacitor ($C=10 \text{ nF}$): While this value might seem low for such a long period, it helps stabilize the cycles and avoids the significant leakage currents often found in large electrolytic capacitors.
- The Resistors ($R_A = 1 \text{ k}\Omega, R_B = 100 \text{ k}\Omega$): By using values in the $\text{k}\Omega$ range, we keep power consumption low while staying well within the LM555's operating limits.

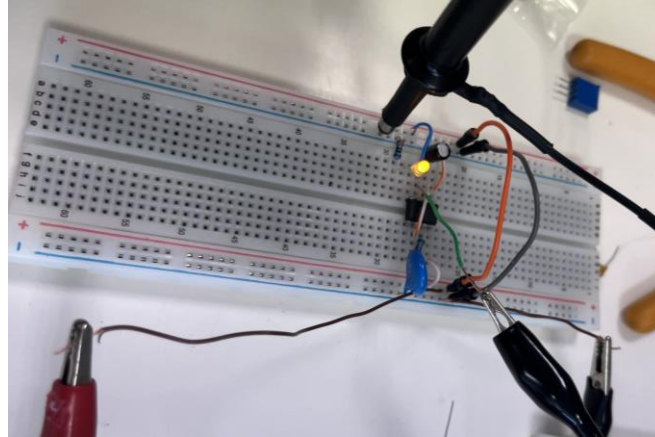
The duty cycle is defined as the ratio of the "high time" (T_r) over the total period (T).

- High time formula: $T_r = 0.693 \cdot (R_A + R_B) \cdot C$
- Low time formula: $T_f = 0.693 \cdot R_B \cdot C$

Theoretically, to get a perfect 50% duty cycle, R_A would need to be zero. However, this isn't feasible because Pin 7 (Discharge) would short-circuit VCC directly to ground during the discharge phase, which would likely fry the IC.

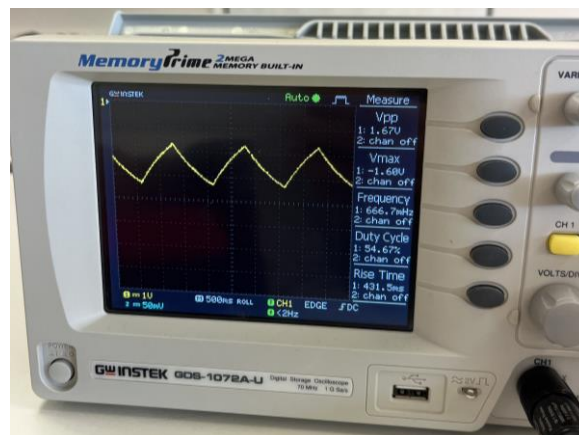
Our strategy was to choose R_B (100 k Ω) to be significantly larger than R_A (1 k Ω). Since $R_B \gg R_A$, then $(R_A + R_B) \approx R_B$, making $T_r \approx T_f$. This allowed us to approach a 50% duty cycle while keeping the component safe.

We used a 10 nF (3 kV) capacitor as the circuit's energy reservoir; its capacity, combined with the resistors, dictates how fast the voltage thresholds (1/3 and 2/3 of VCC) are reached. R_A is set at 1 k Ω to limit current during discharge, while R_B (100 k Ω) influences both the charge and discharge phases. Finally, an LED is connected to Pin 3 through a current-limiting resistor (R_L) to provide a visual confirmation of the clock oscillations.



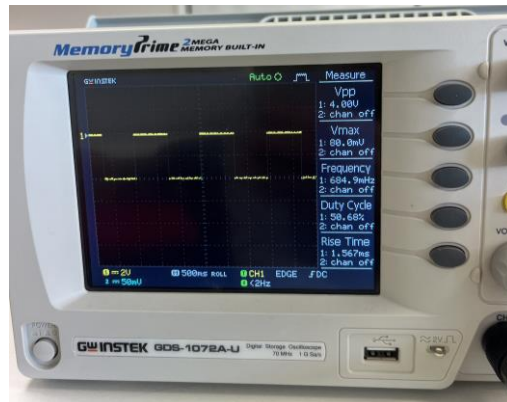
2 – Signal Analysis

When probing the signal across the capacitor, we observed a sawtooth (triangular) waveform charging and discharging between $1/3V_{CC}$ and $2/3V_{CC}$. This signal is what triggers the 555's internal comparators and confirms the capacitor is cycling correctly.

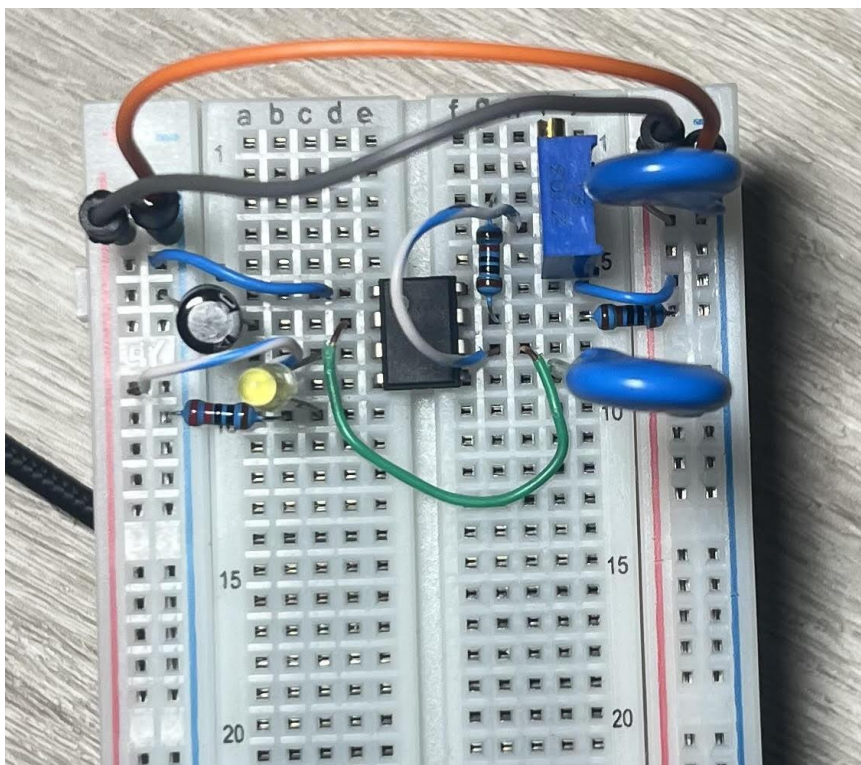


On the output pin (Clock_out / LED), we get a clean square wave. As expected, the period is about 1.5s, resulting in a very low frequency that allows the LED to blink clearly. With R_B being much larger than R_A ,

the duty cycle is nearly 50%. Since $T_r \approx 0.693 \cdot R_B \cdot C$ and $T_f \approx 0.693 \cdot R_B \cdot C$, the high and low times become almost identical, as R_A becomes negligible in the calculation.



3 - Second Circuit Version with Adjustable Frequency



In the second version of the clock circuit, the fixed resistor $R_B = 100\text{k}\Omega$ was replaced by a combination of a **$33\text{k}\Omega$ fixed resistor in series with a $100\text{k}\Omega$ potentiometer**, allowing the output frequency to be adjusted experimentally. In this configuration, the effective resistance becomes $R = 33\text{k}\Omega + R_{\text{pot}}$, where R_{pot} varies from 0 to $100\text{k}\Omega$. As a result, the oscillation period is no longer fixed, but can be tuned according to the relation $T = 0.693 \cdot (R_A + 2R_B) \cdot C$, while the frequency is given by $f = 1.44 / (R_A + 2R_B) \cdot C$. The addition of the fixed $33\text{k}\Omega$ resistor ensures that the total resistance never becomes too small, thus preventing the frequency from rising excessively and maintaining safe and stable operation of the LM555. At the same time, since $R_A = 1\text{k}\Omega$ remains much smaller than the effective R , the duty cycle stays close to 50% throughout the adjustment range. This second implementation is therefore more practical for the laboratory setup, since it preserves the nearly symmetrical square wave of the first version while also providing the ability to fine-tune the blinking rate of the LED and the clock frequency of the circuit.

Capacitor's Role

In addition to the main timing capacitor, two auxiliary **ceramic capacitors marked “103 3 kV”** were used in the practical implementation of the circuit. The code **103** corresponds to a capacitance of **10 nF ($0.01\text{ }\mu\text{F}$)**, while the **3 kV** marking indicates their maximum voltage rating, which is far above the operating voltage of this 5 V circuit and therefore provides a large safety margin. These capacitors were added for **noise suppression and circuit stabilization**. One of them is connected to the **control-voltage pin (pin 5)** of the LM555 and to ground, in accordance with the recommended astable configuration. Its role is to filter unwanted disturbances and stabilize the internal reference thresholds of the comparators,

preventing false triggering due to electrical noise. The second ceramic capacitor acts as a **decoupling/bypass capacitor** for the power rails, reducing high-frequency supply noise and transient disturbances generated during switching. In practice, the presence of these capacitors makes the oscillation more stable, improves waveform quality, and prevents erratic behavior of the circuit when observed on the oscilloscope or when connected to the rest of the digital system.

Soldering Part

After verifying the correct operation of the circuit on the breadboard and confirming the expected waveforms with the oscilloscope, the next step is its transfer to a **perfboard** for permanent implementation. This stage is necessary in order to transform the temporary prototype into a more robust and reliable hardware circuit suitable for laboratory use. Unlike the breadboard, where many electrical connections are internally provided by the board structure, the perfboard consists of isolated pads, meaning that all required connections must be recreated manually using soldered wires and component leads. For this reason, special attention must be given to component placement, routing of the power and ground lines, and the overall layout of the circuit, so that the final implementation remains electrically correct, stable, and easy to debug.

Conclusion

In this laboratory exercise, an analog clock circuit based on the **LM555 timer in astable mode** was successfully designed, implemented, and experimentally tested. The theoretical study of the circuit clarified the internal operating principle of the LM555, the role of the comparator thresholds at $(1/3)VC$ and $(2/3)VC$, and the way in which the repeated charging and discharging of the capacitor produces a periodic square-wave output. The mathematical analysis also showed how the values of $R1, R2, C$ determine the period, frequency, and duty cycle of the generated signal.

In the practical part, the circuit was first assembled and tested on a breadboard, where the expected waveforms were confirmed using the oscilloscope. The output at pin 3 was found to be a stable square wave, while the capacitor voltage followed the expected charging–discharging behavior between the theoretical threshold levels. The use of a small R_A *and a much larger R_B* allowed the duty cycle to remain close to 50%, producing an almost symmetrical clock pulse suitable for digital timing applications. In the improved version, the replacement of the fixed R_B resistor with a series combination of a $33\text{ k}\Omega$ resistor and a $100\text{ k}\Omega$ potentiometer provided the additional advantage of adjustable frequency, making the circuit more flexible for laboratory use.

Overall, the implementation verified both the theoretical predictions and the practical reliability of the LM555 astable oscillator. The final circuit can therefore be used effectively as an external clock source for the subsequent stages of the digital systems laboratory project.